

Question (3 Marks)



SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	Slot	B
Course Title	Cloud Computing and Big Data Analytics	Assessment	CO4 AT1
Assessment Title	Big Data Analytics Class Test	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	Total Marks	30

Student Details

Student Name	T.PRAKEERTHI	Register Number	192524279
Class / Section	B.Tech 2 nd year	Slot	B
Capstone Title	MediQueue AI: Hospital OP Queue and No-Show Prediction App	Submission Date	19/08/2026
GitHub Repository Link	https://github.com/tarigondaprakeerthi/CSA1512-cloud-computing-Assessment.git	Google Classroom Status	

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.

Question (3 Marks)

- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check
Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?
Coverage	6	All required parts of the question are answered.	Have I answered every part?
Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I T.PRAKEERTHI(192524279) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: I T.PRAKEERTHI

Question (3 Marks)

and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

Ans: Big Data

Big Data refers to datasets whose size, speed, variety, or complexity exceeds the capability of conventional data-processing tools.

Traditional approaches such as spreadsheets and single-server databases are often insufficient because:

- **Storage limits:** They cannot efficiently store extremely large datasets.
- **Slow processing:** Queries and analysis become very slow as data grows.
- **Concurrency issues:** Handling many users and data requests at the same time becomes difficult.
- **Schema rigidity:** Traditional databases often require structured and predefined data formats.
- **Limited scalability:** A single server has limited CPU, memory, and storage capacity.

Example: A cloud-based shopping application may generate millions of transactions, clickstream events, app logs, and images every day. Processing all this data on one server can become slow and expensive.

Therefore, Big Data environments require distributed storage and parallel processing, where data and computation are spread across multiple machines to handle large volumes of data efficiently.

Question (3 Marks)

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

Ans: 5Vs of Big Data

1.**Volume:** Refers to the huge amount of data generated and stored. It affects storage capacity and scalability.

2.**Velocity:** Refers to the speed at which data is generated, collected, and processed. It affects real-time processing requirements.

Example: In a smart traffic system, continuous vehicle location updates are generated every second. The system must process these updates quickly to monitor traffic.

3.**Variety:** Refers to different types of data such as text, images, videos, audio, and sensor data. It affects data storage and processing methods.

4.**Veracity:** Refers to the accuracy, reliability, and quality of data. Poor-quality data can affect analytics and decision-making.

5.**Value:** Refers to the useful information or benefits obtained from data. It affects analytics and business decisions.

Example: A smart traffic system can analyze vehicle data to predict congestion and suggest better routes.

Thus, the 5Vs help organizations decide how data should be stored, processed, managed, and analyzed efficiently

T3

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.

Question (3 Marks)

3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

Ans: Types of Data

Structured Data: Has a fixed schema and is stored in tables.

Example: Customer and transaction records.

1. **Semi-structured Data:** Has flexible keys or tags.
Example: JSON API responses and log files.
2. **Unstructured Data:** Has no fixed structure.
Example: Images, videos, PDFs, and emails.

Example: A cloud shopping app may store transactions as structured data, API responses as semi-structured data, and product images as unstructured data.

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

Ans: Role of Distributed File Systems

A Distributed File System (DFS) stores very large datasets by splitting them into blocks and distributing them across multiple machines.

- Provides scalability and fault tolerance.
- Supports parallel access and large-scale batch processing.
- HDFS is a common example. Its data blocks can be processed in parallel using MapReduce or Spark.

Thus, DFS enables efficient storage and processing of Big Data.

Question (3 Marks)

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

Ans: Big Data Security Concerns

Unauthorized Access: Attackers may access sensitive data. Control: Use role-based access control (RBAC).

1. **Data Leakage:** Sensitive information may be exposed or stolen. Control: Use encryption and data masking.
2. **Privacy Violation:** Combining data from multiple sources may reveal personal information.
3. **Control:** Use anonymization and proper monitoring.

Thus, Big Data requires strong security and governance because data comes from many different sources.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Question (3 Marks)

Student Response

Ans: Data Appliances and Integration Tools

Data appliances are specialized hardware/software systems optimized for data storage, processing, and analytics.

Integration tools move, transform, and synchronize data between different systems. Examples include ETL/ELT tools, API connectors, message queues, and schedulers.

Big Data workflow:

Source Data → Ingestion → Cleaning/Transformation → Storage → Processing → Dashboard/Decision

Thus, they help organizations efficiently collect, process, and analyze data from multiple sources.

7

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

Ans: Challenges in Big Data

1. **Data Quality:** Data from multiple sources may be incomplete or inaccurate, leading to wrong analysis and business decisions.
Mitigation: Use data cleaning and proper data governance.
2. **Infrastructure Cost:** Storing and processing huge datasets requires expensive infrastructure, increasing business costs.
Mitigation: Use **cloud scaling** and pay only for required resources.

Thus, managing data quality and infrastructure costs is essential for successful Big Data implementation.

Question (3 Marks)

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

Ans: Big Data Analytics in Healthcare

Data Sources: Patient records, medical reports, hospital logs, and wearable devices.

Use of Analytics: Big Data analytics can predict diseases, identify high-risk patients, and optimize hospital resources.

Outcome: It helps reduce patient waiting time, improve treatment decisions, and provide better healthcare services.

9

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Business Drivers for Big Data

1. **Cost Reduction:** Helps identify unnecessary expenses, requiring efficient data analysis and cloud resource management.
2. **Personalization:** Uses customer behavior data to provide personalized recommendations, requiring data tracking and analytics.

Question (3 Marks)

3. **Risk Control:** Detects fraud and unusual activities, requiring real-time monitoring and anomaly detection.

Thus, Big Data adoption should provide measurable business value through better decisions, lower costs, and improved customer experience.

Checklist Item	Status	Remarks
Student details completed.	Done / Pending	DONE
All 10 questions answered.	Done / Pending	DONE

Question (3 Marks)

Examples are connected to cloud/capstone/domain context.	Done / Pending	

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

Importance of Big Data Analytics in Decision Making

Big Data analytics converts large amounts of raw data into useful patterns, predictions, alerts, and recommendations for better decisions.

- **Descriptive:** What happened?
- **Diagnostic:** Why did it happen?
- **Predictive:** What may happen?
- **Prescriptive:** What should be done?

Example: A bank can use analytics to detect unusual transactions and generate fraud alerts.

Thus, Big Data analytics makes decisions faster, evidence-based, and scalable.

Submission Checklist

Security/privacy points are included wherever relevant.	Done / Pending	DONE
Document exported as PDF if requested.	Done / Pending	DONE
GitHub link shared through Google Classroom.	Done / Pending	DONE